

CONCEPT NOTE

AI-Based Labor-Market Skills Demand Intelligence System

SkillSense

A Predictive Labor-Market Intelligence and Education/Training Alignment Platform for Rwanda

Prepared by

SkillSense Team

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1. Executive Summary

SkillSense is a proposed AI-powered labor-market intelligence platform designed to help understand how employment demand changes over time, which job roles are growing or declining, which skills are associated with those roles, and how favorable the employment outlook for different roles may be in the near future.

The project will use historical labor-market information as its main evidence base. This may include records or summaries showing job vacancies and roles from public and private organizations, applications for different roles, and employment outcomes where available. Current and recent job postings will provide an additional up-to-date signal of labor-market demand.

The system will not attempt to predict whether every individual student will obtain employment. Instead, it will provide a role-level **Employability Outlook** based on observed demand, employment evidence where available, and forecasts for 6, 12 and 24 months. This makes the project measurable and aligned with the data that can reasonably be obtained.

2. Problem and Rationale

Labor-market needs change as sectors grow, technologies evolve and economic priorities change. Education and training institutions therefore need evidence about which roles are being demanded and which skills are becoming more important. A historical view is especially valuable because it allows current demand to be interpreted as part of a longer trend rather than as a single snapshot.

SkillSense will organize this evidence around a simple relationship:

**SECTOR → JOB ROLE → REQUIRED SKILLS → DEMAND TREND → FUTURE
OUTLOOK**

3. Main Purpose

To build a Rwanda-focused platform that analyzes historical and current labor-market information to identify role and skill demand, detect emerging roles, forecast future role demand, and translate the findings into evidence-based employability, career, education and training insights.

4. How the System Will Work

Stage	What Happens
1. Historical labor-market data	Government and other authorized sources provide information about vacancies, roles, applications, employment outcomes and related historical indicators.
2. Current labor-market data	Authorized job postings from relevant sources provide recent and continuously updated signals of employer demand.
3. Data preparation	The system cleans and organizes the information, removes duplicates, standardizes job roles and skills, and organizes records by sector, location and time.
4. Sector-specific analysis	The labor market is analyzed sector by sector rather than treating all sectors as one market. ICT/Computer Science is the first implementation sector.
5. Intelligence and forecasting	The system measures role demand, identifies growing/emerging roles and associated skills, and forecasts role demand for 6, 12 and 24 months where sufficient historical data exists.
6. Decision support	The results are presented to career/training advisors, education planners and labor-market analysts through dashboards and a grounded conversational AI interface.

5. Data Required from the Government / Data Custodian

The project team is particularly interested in existing historical labor-market datasets. The purpose of this section is to show the types of information that would make the system more accurate; the data custodian can determine which of these already exist and what can be shared under applicable rules.

Data Category	Preferred Information	Why It Matters
Historical vacancies / job records	Job role/title, sector, location, date, requirements/skills, and other available vacancy information.	Understand how demand for roles changed over time.
Application records	Role applied for, application date, relevant qualification/education information, location and application outcome where available.	Understand application patterns and competition.

Data Category	Preferred Information	Why It Matters
Employment outcomes	Whether an applicant became employed, employment date where available, role/sector/location of employment where available.	Relate demand to actual employment outcomes.
Historical labor-market summaries	Employment/unemployment indicators, occupation or sector trends and other dated labor-market statistics where available.	Provide historical context and support validation.
Official classifications	Occupation, sector, geographic and education/qualification classifications and definitions.	Keep the analysis consistent with official categories.

5.1 Preferred Historical Coverage

If reliable historical records are available across a long period, they may allow the project to study how the labor market changed during earlier national development periods, including the period associated with Vision 2020, and compare those patterns with the current situation. The same evidence can support forward-looking analysis toward the Vision 2050 horizon. The project will not assume that all historical years are directly comparable; definitions, coverage and data quality will first be assessed.

6. Current Job-Posting Data

Current job postings will complement the historical government data by providing a more recent view of employer demand. These records may be obtained through authorized APIs, CSV datasets or permitted web scraping. The location contained in each job posting will be used for geographic analysis; a separate geographic dataset is therefore not required for the core labor-market analysis.

Preferred Job-Posting Information	Purpose
Job title / role	Identify and standardize occupations.
Job description and requirements	Extract and standardize required skills.
Posting date	Measure recent demand and update trends.
Location	Analyze where roles are demanded.
Sector / industry, if available	Support sector classification and validation.
Experience / education requirements, if available	Profile roles and support employability analysis.
Source	Track data coverage and provenance.

7. University and Training Information

University curricula and training-provider information are not expected to be supplied as part of the government labor-market dataset. They will be entered into SkillSense by authorized users.

Education planners or authorized users can upload university bulletins, curriculum documents and programme information. Training providers can submit their own training programmes and the skills they teach. SkillSense will then compare these user-provided skills with the skills demanded by the labor market.

Information	How It Enters the System	Purpose
University programmes and curricula	Uploaded by authorized education users	Compare taught skills with market demand.
Courses/modules and descriptions	Uploaded as part of curriculum information	Identify skills covered by programmes.
Training programmes	Entered by training providers or authorized users	Identify available learning opportunities.
Skills taught	Entered/uploaded by training providers and education users	Compare training supply with demanded skills.

8. Geographic Intelligence

Geographic analysis will be derived directly from the location recorded in historical labor-market records and job postings. The system will normalize available locations to the finest reliable administrative level supported by the data. This allows the same role-demand analysis to be viewed by location without requiring a separate geographic data source.

ROLE × SECTOR × LOCATION × TIME

9. Sector-Specific Intelligence

SkillSense will not treat the entire labor market as one homogeneous dataset. Labor-market behavior, occupations and skill requirements differ substantially between sectors. Therefore, sector identification and filtering will occur before sector-specific demand and forecasting analysis.

Sector	Illustrative Roles and Skills	Approach
ICT / Computer Science	Software, data, cybersecurity, cloud, IT support and related roles.	First complete implementation and validation.
Finance	Banking, accounting, finance, risk, insurance and related roles.	Add after sufficient data is available.
Healthcare	Clinical, laboratory, public-health, administration and related roles.	Add after sufficient data is available.
Agriculture / Agribusiness	Agricultural, agribusiness, extension and related roles.	Add after sufficient data and classification support.
Construction / Infrastructure	Engineering, construction, infrastructure and project roles.	Add after sufficient data is available.

The system architecture will remain sector-agnostic so that additional sectors can be introduced without rebuilding the core platform. The number of sectors will be determined by data availability and quality rather than by an arbitrary requirement.

10. What SkillSense Will Produce

Output	Meaning
Role demand	How frequently a role is observed in the available labor-market data.
Role trend	Whether demand for a role is growing, stable or declining.
Emerging role	A role showing sufficiently strong and sustained growth according to defined analytical criteria.
Role skill profile	The skills most commonly associated with a particular role.
Skill trend	How the prevalence of a skill changes within relevant roles over time.
Geographic demand	Where particular roles are being demanded.
Forecast	Expected role-demand direction/value for 6, 12 or 24 months where supported by data.
Employability Outlook	A role-level assessment of how favorable employment prospects appear based on demand and available employment-outcome evidence.
Curriculum alignment	How closely a programme's taught skills correspond to skills demanded by relevant roles.
Training alignment	How closely available training offerings correspond to current and emerging market needs.

11. System Users

User	What They Will Use SkillSense For
Career & Training Advisor	Explore promising roles, required skills, employability outlooks and learning/training recommendations.
Education / Curriculum Planner	Compare programmes and curricula with market-demand skills and identify areas for review.
Labor Market Analyst	Analyze roles, skills, sectors, locations, historical trends, employment outcomes and forecasts.
Administrator	Manage data sources, users, taxonomies, uploads, data quality and system configuration.

12. AI and Forecasting Approach

The project will use AI and statistical techniques where they provide measurable value. The exact algorithms will be selected only after the available data has been inspected. The team will not assume that one algorithm should be used for every sector or every analytical problem.

The primary prediction target will be future demand for job roles. Skills will be analyzed mainly as attributes of roles and as inputs to recommendations. Where sufficient application and employment-outcome data are available, these records will be used to strengthen and validate the Employability Outlook.

Every forecast will state the data cutoff date and forecast period. For example: *'Data through July 2026; 12-month outlook: August 2026 – July 2027.'* If there is insufficient historical evidence for a particular sector, role or forecast horizon, the system will report that limitation rather than produce an unsupported prediction.

13. Conversational AI

A conversational AI interface may be integrated using a service such as the Gemini API. It will act as a natural-language interface and explanation layer over SkillSense's structured results. For example, a user may ask what skills are associated with an emerging role in ICT, and the assistant can explain the system's calculated results.

The conversational model will not be the source of the underlying labor-market statistics or forecasts. Responses will be grounded in SkillSense data and analytical outputs.

14. Implementation Scope

Phase	Focus
Phase 1	Data discovery, access assessment and data-quality evaluation.
Phase 2	ETL pipeline and common labor-market data model.
Phase 3	Complete ICT labor-market intelligence and geographic analysis.
Phase 4	Role forecasting and Employability Outlook.
Phase 5	University curriculum and training-provider alignment.
Phase 6	Role-based dashboards and conversational AI.
Phase 7	Validation and preparation for additional sectors.

15. Information Requested from the Data Custodian

The project team respectfully requests assistance in identifying whether the institution has existing datasets containing some or all of the following information. The team does not require personally identifying information.

- Historical labor-market records covering vacancies, occupations/roles, sectors, locations and dates.

- Where available, anonymized records of applications for roles and their outcomes.
- Where available, employment outcomes associated with applications or employment programmes.
- Historical employment/unemployment or occupation/sector summaries that can provide dated context.
- Official occupation, sector, geographic and education/qualification classifications.
- For each dataset: available years/date range, approximate number of records, field definitions, geographic coverage and update frequency.

Any shared data would be used only within the permissions and requirements established by the relevant data custodian. Direct identifiers such as names, national identification numbers, phone numbers and email addresses are **not required** for the analytical objectives of the project.

16. Expected Value

- Provide a clearer evidence-based view of changing role and skill demand.
- Identify emerging roles and the skills associated with them.
- Provide short- and medium-term labor-market outlooks rather than relying only on historical rankings.
- Connect labor-market demand with observed employment outcomes where suitable government data is available.
- Help career and training advisors identify relevant learning priorities.
- Help education planners identify potential curriculum-market alignment gaps.
- Provide labor-market analysts with sector and geographic intelligence.
- Create a reusable foundation that can expand from ICT to other sectors as reliable data becomes available.

17. Important Interpretation and Data Limitations

- Job postings represent observed recruitment demand from the sources included in the system and may not capture every vacancy in Rwanda.
- Historical government records may use different definitions or classifications across periods; comparability will therefore be assessed before modeling.
- A sector-specific model or forecast will only be produced where sufficient data exists.
- The Employability Outlook is a role-level analytical outlook, not a guarantee that a particular person will obtain employment.
- University and training recommendations support human decision-making and do not automatically determine curriculum changes.

18. Conclusion

SkillSense is designed as a focused and implementable labor-market intelligence project. Its core evidence will come from historical labor-market information and current job postings, allowing the

system to connect how the labor market has performed with what employers are demanding now and what role demand may look like in the next 6, 12 and 24 months.

The project will begin with ICT/Computer Science and will analyze sectors independently before expanding to other priority sectors. Geographic intelligence will be derived from the locations already contained in labor-market records and job postings. University curricula and training information will be supplied through the platform by authorized users rather than being required as part of the government labor-market dataset.

The immediate collaboration need is therefore straightforward: identify which historical labor-market, application, employment-outcome and official classification datasets are available, their coverage and definitions, and the conditions under which they can be used for this project. The final AI models and prediction methods will be selected after the actual data is assessed.

Respectfully submitted,

The SkillSense Project Team

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